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Escuela de Ingeniería
Profesor Denis Parra



DLCRec: A Novel Approach for Managing *diversity* in LLM-Based Recommender Systems

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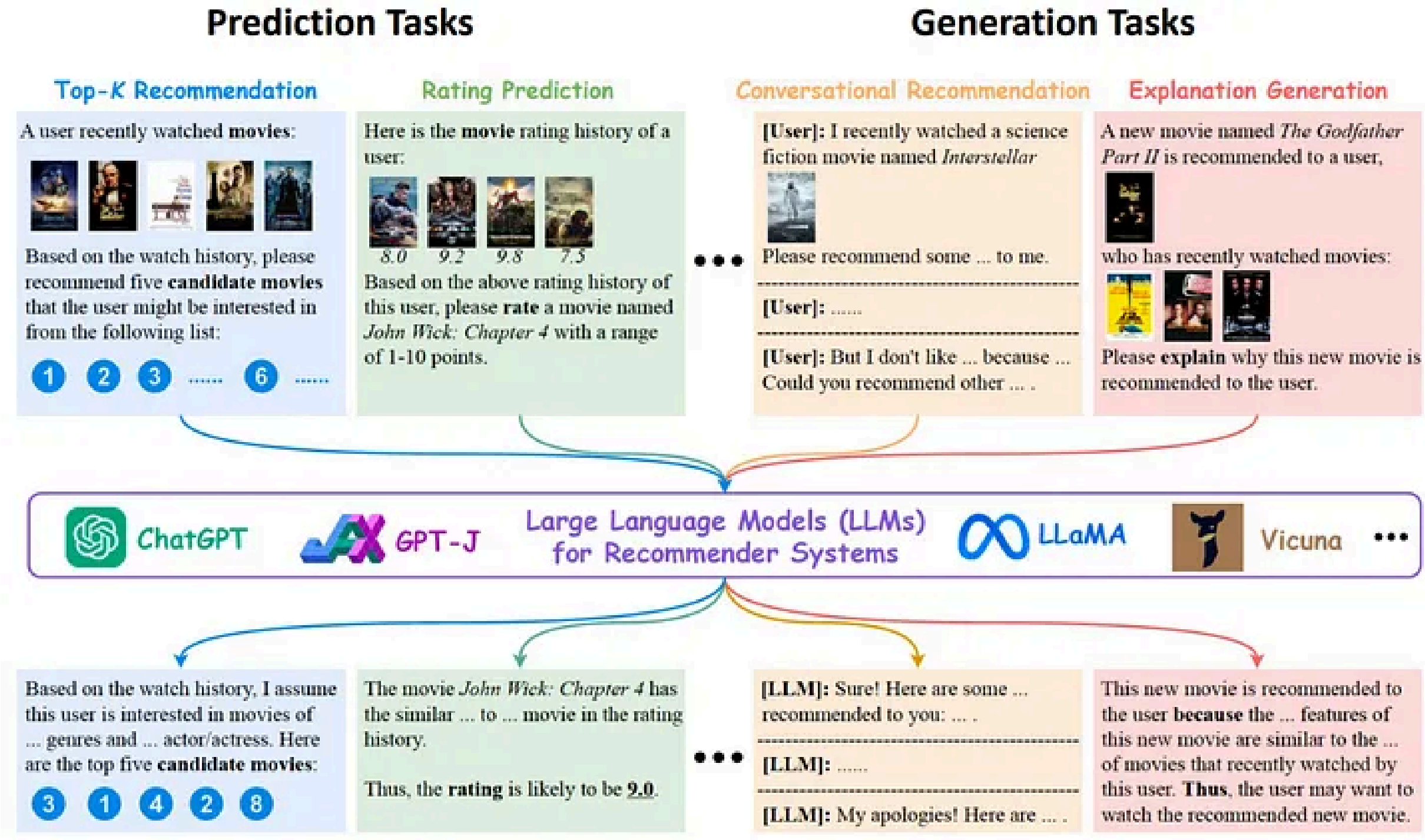
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Catalina Anusch José Villalobos Anibal Villegas

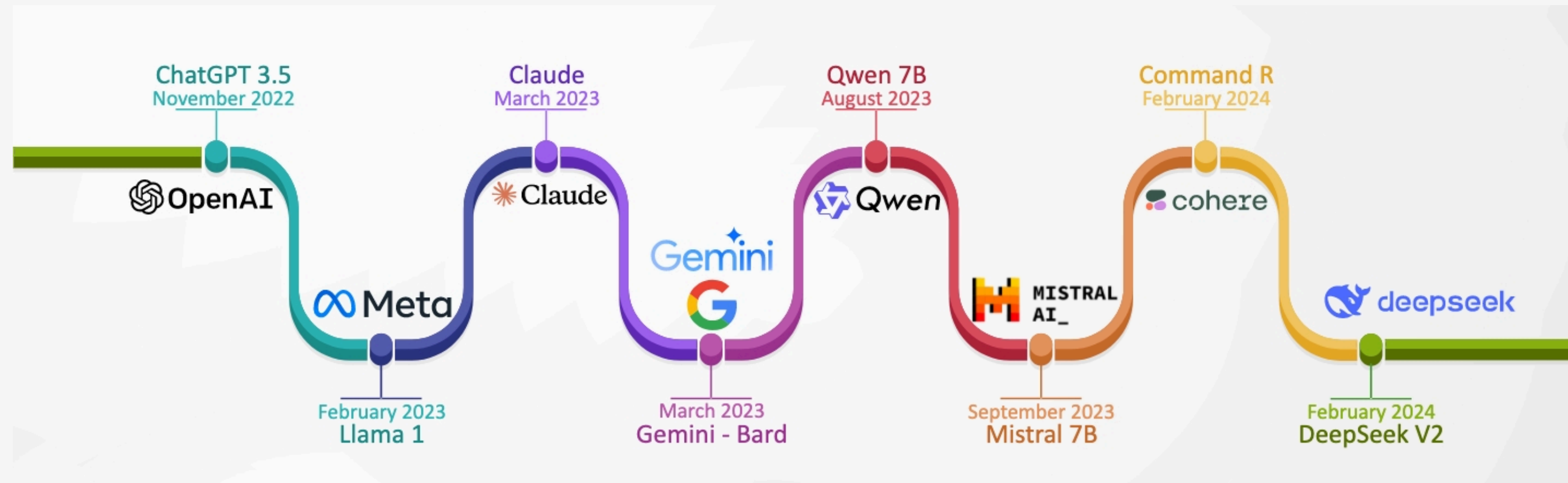
1. Contexto de la Investigación

LLMs

Modelos entrenados con conjuntos de datos masivos de texto que comprenden y generan **lenguaje humano**



1. Contexto de la Investigación



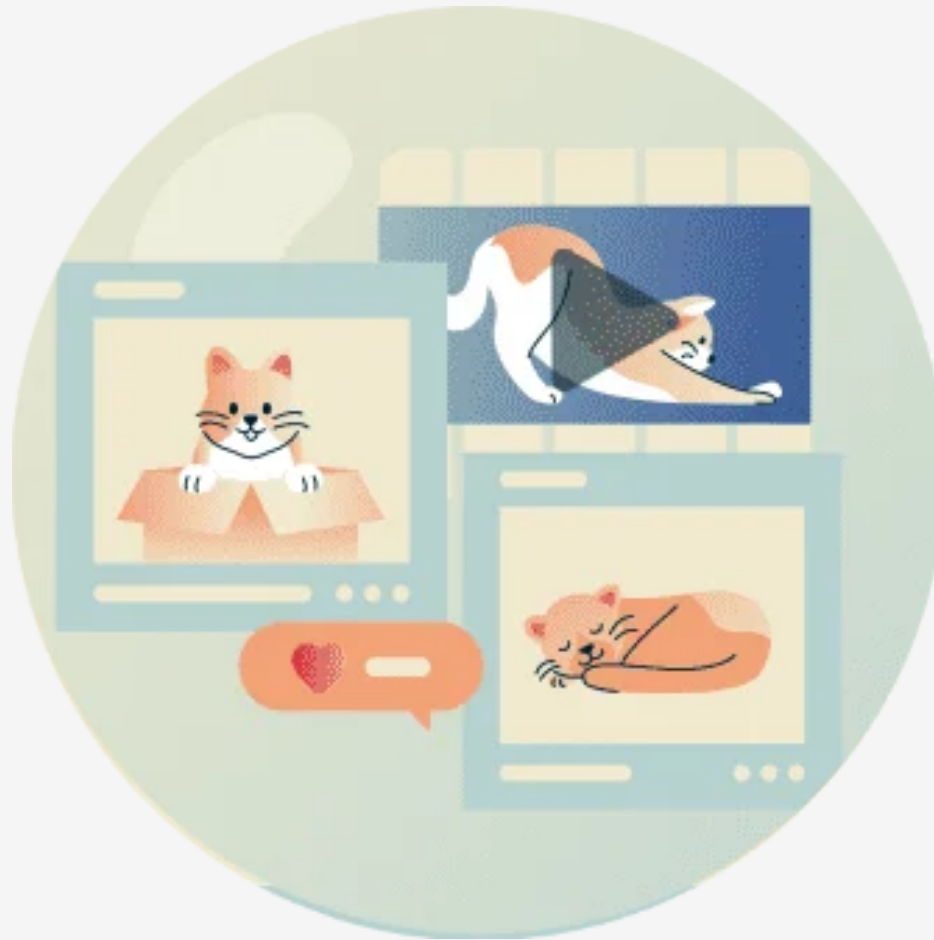
LLMs + RecSys = Gran captura de **contenido semántico**

Few-shot prompting

Tan solo un par de ejemplos en la instrucción...

1. Contexto de la Investigación

Filter bubble



Contenido personalizado

Usuarios atrapados en contenido demasiado similar a su historial, ahora no solo se requiere precisión, sino que también...

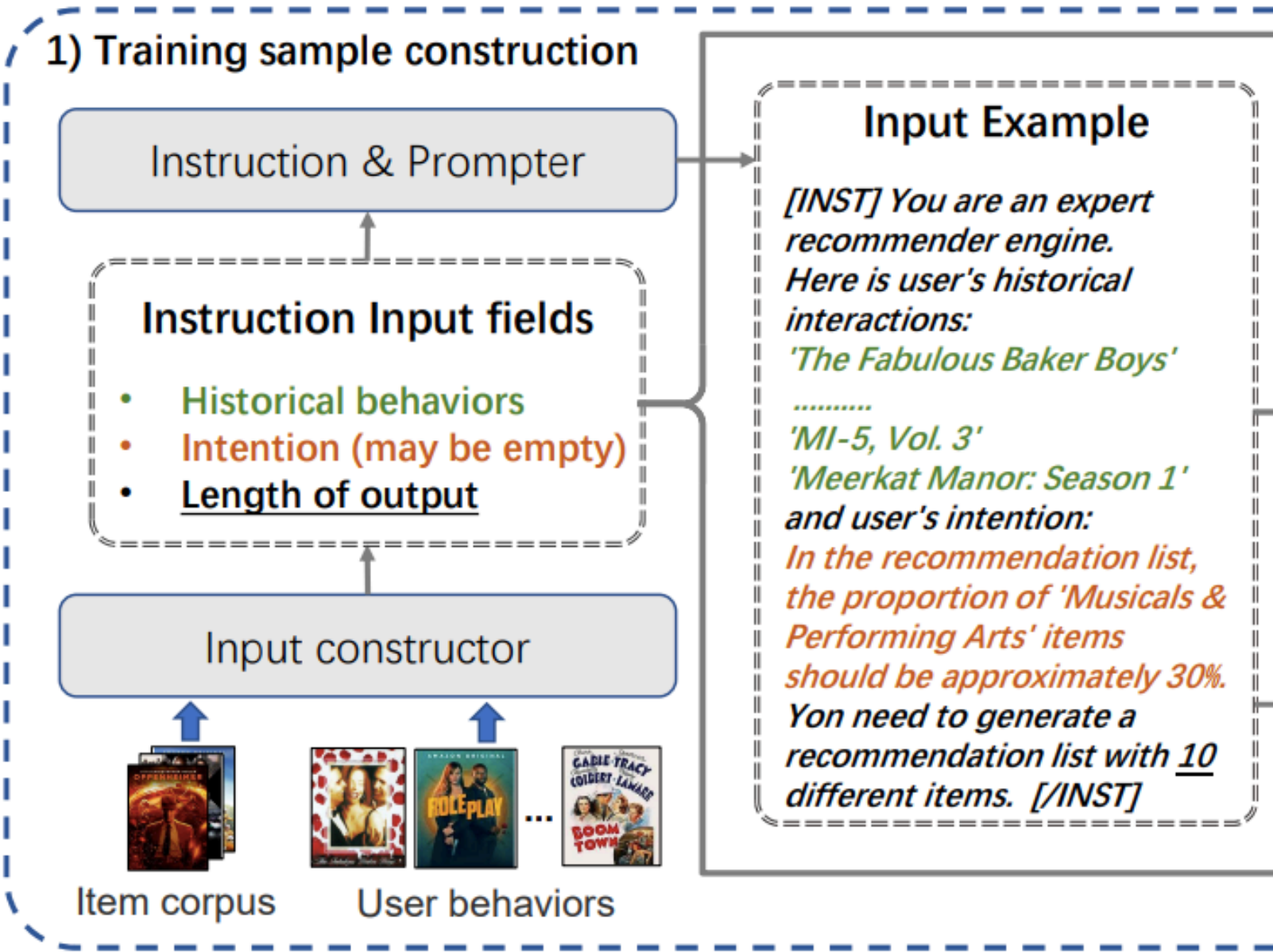
Diversidad

Echo chamber



Creencias/gustos preexistentes

Sistemas recomendadores controlables



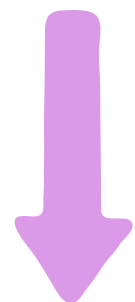
Pero, ¿cómo abordar complejidad?

2. Problema de Recomendación

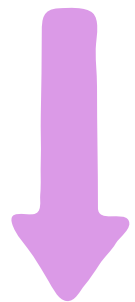
Sea un usuario u con historial de interacciones

$$H_u = \{(i_1, g_1), (i_2, g_2), \dots, (i_n, g_n)\}$$

donde $i_k \in \mathcal{I}$ es un ítem y $g_k \in \mathcal{G}$ su género.

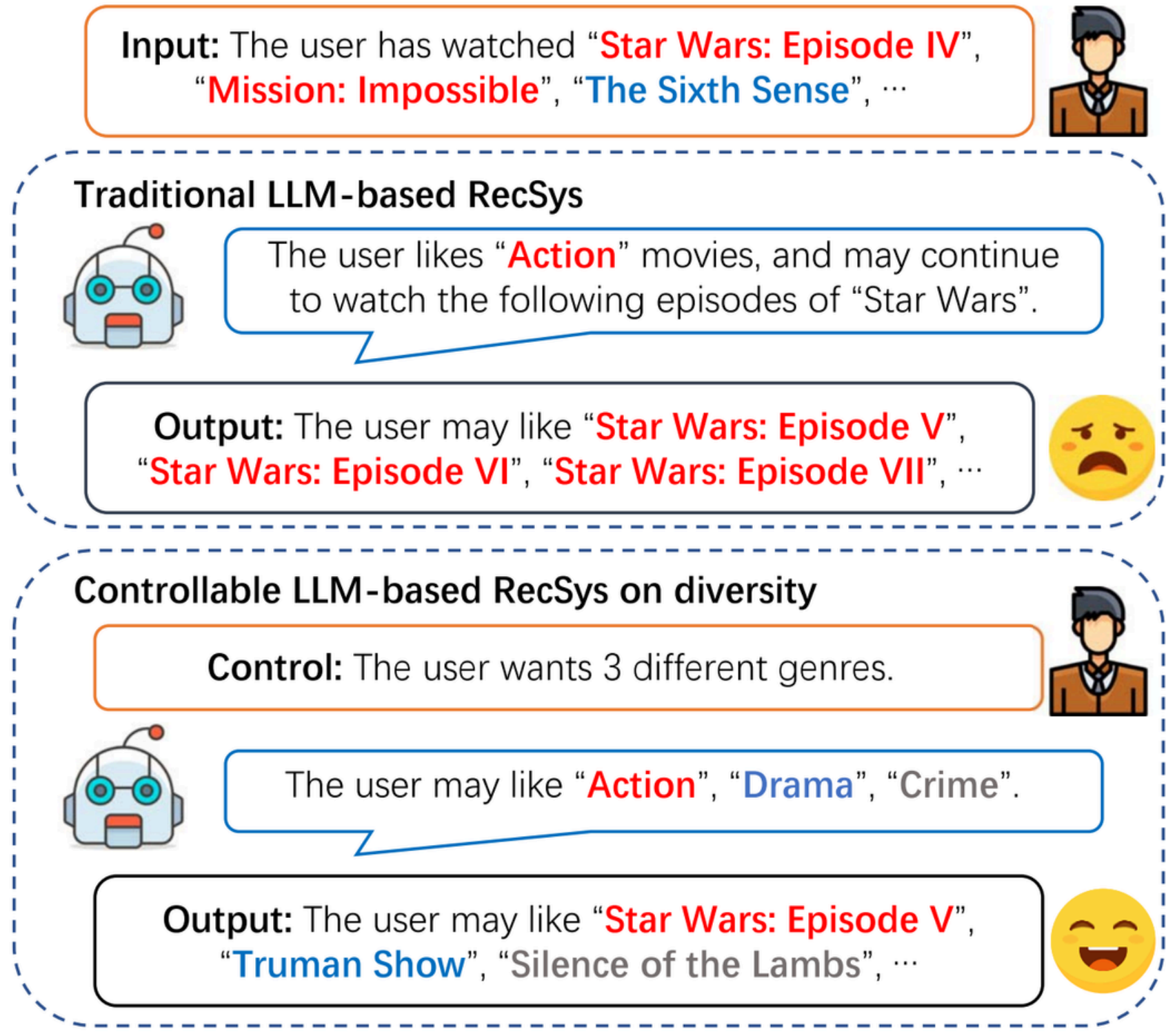


Generar una lista $\hat{L} = [\hat{i}_1, \hat{i}_2, \dots, \hat{i}_K]$



Número de control

$N_c \in \mathbb{Z}^+$: Cantidad de géneros distintos deseados



3. Contribución

Nuevo Paradigma

Control de **grano fino** sobre la diversidad

Nuevo Framework

DLCRec: **Diversity**
oriented LLM-based
Controllable
Recommender

Genre Predicting (GP)

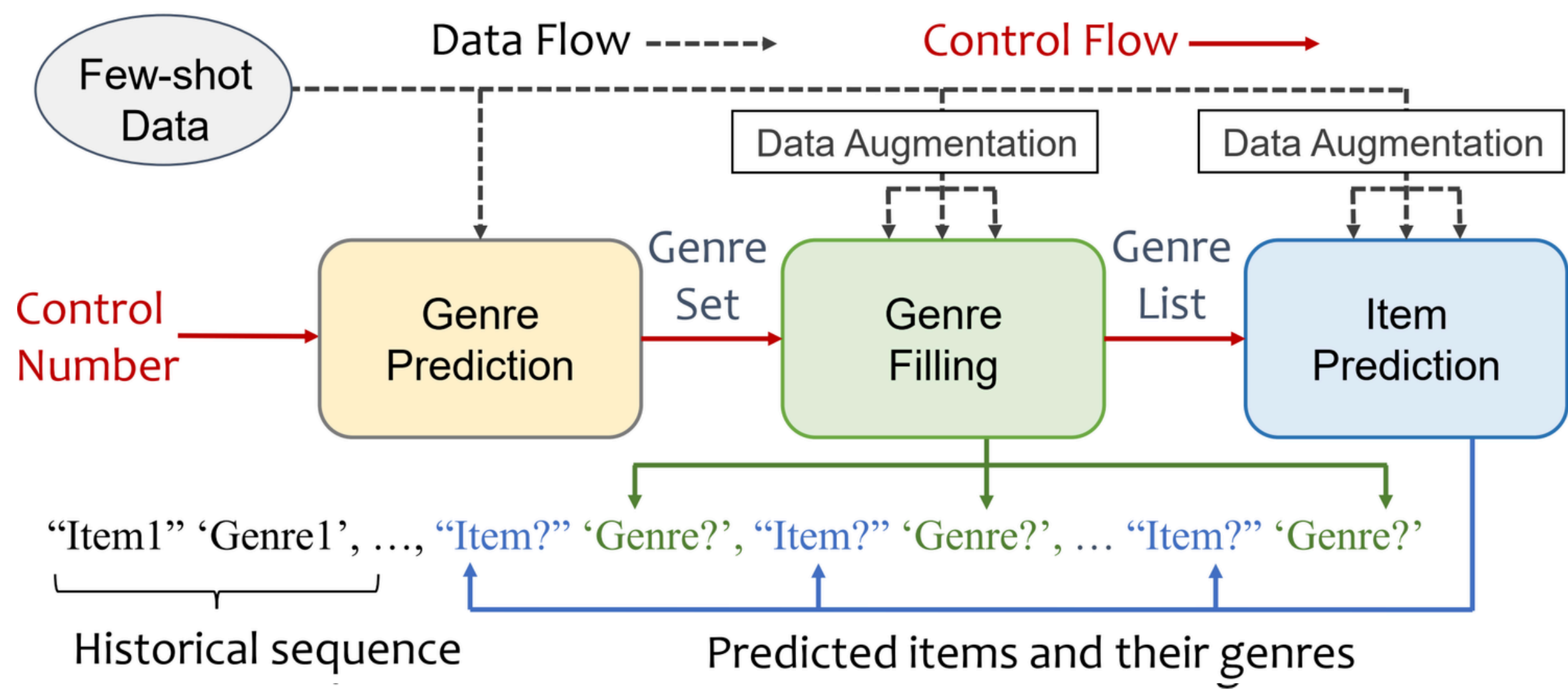
Genre Filling (GF)

Item Predicting (IP)

Data Augmentation

Técnicas pioneras para
inyectar ruido y alterar
distribuciones

3. Contribución



Instruction tuning

Se entrena al LLM con pares de **instrucción → respuesta esperada**



Instrucciones de una sola tarea fallan en capturar complejidad

RLFT: Reinforcement learning fine-tuning

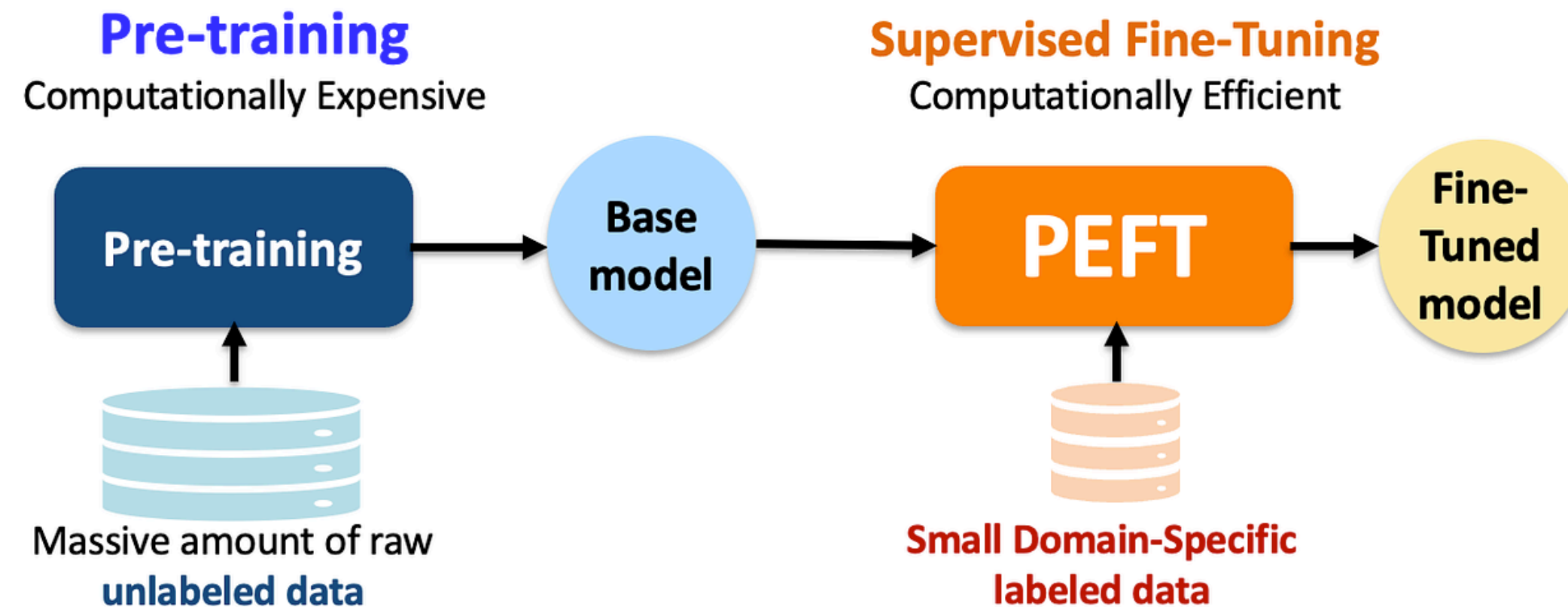
Señales de recompensa para refinar el comportamiento del modelo después del fine-tuning supervisado



Intensivos en datos + problemas de convergencia

PEFT: Parameter Efficient Fine-Tuning

Solo modifica una **pequeña parte** de los parámetros dentro de un modelo



Chain of Thought (CoT)

Técnica de prompting avanzado que consiste en pedir razonamiento **paso a paso** antes de dar la respuesta final

Without chain-of-thought

Answer the following
yes/no question.



yes

Can you write a whole
Haiku in a single tweet?

With chain-of-thought

Answer the following yes/no question
by reasoning step-by-step.



Can you write a whole Haiku in a
single tweet?

A haiku is a japanese
three-line poem.
That is short enough
to fit in 280
characters. The
answer is yes.

5. Detalle de la solución

Función objetivo
del **instruction tuning**
para una tarea T

$$\max_{\Phi} \sum_{(x,y) \in \mathcal{Z}} \sum_{t=1}^{|y|} \log P_{\Phi}(y_t \mid x, y_{<t})$$

- Parámetros
del LLM que
se optimizan
- 

 - x : Instrucción o prompt de entrada
 - y : Respuesta esperada (label)
 - y_t : Token que el modelo está intentando predecir en el paso t
 - $y_{<t}$: Todos los tokens generados antes de y_t
 - $P_{\Phi}(y_t \mid x, y_{<t})$: Probabilidad que el modelo asigna al token y_t dado el contexto

5. Detalle de la solución

Función objetivo
del **instruction tuning**
para una tarea T

$$\max_{\Phi} \sum_{(x,y) \in \mathcal{Z}} \sum_{t=1}^{|y|} \log P_{\Phi}(y_t \mid x, y_{<t})$$

x : "Dado que el usuario ha visto Star Wars (Acción) y Mission Impossible (Acción)... predice los próximos 3 géneros"

y : "Acción, Drama, Comedia"

y_t : $y_3 = \text{"Comedia"}$

$y_{<t}$: $y_{<3} = \{\text{"Acción"}, *, *\}$

$P_{\Phi}(y_t \mid x, y_{<t})$: Probabilidad de generar "Comedia" dado que ya se generó "Acción,"

1: Genre Predicting (GP)

(a) Genre predicting (GP)
Instruction: Given a list of movies and their corresponding genres the user has watched before, please provide the 3 most likely genres in the future recommendation list. Output the genres only, without movie names, explanations, or numbers. The output format is "Genre1, Genre2, ..., Genre3".
Input: The user has watched the following movies with their corresponding genres in "()" before: "Star Wars: Episode V - The Empire Strikes Back (1980)" (Action), "Mission: Impossible (1996)" (Action), ..., "Stargate (1994)" (Action).
Output: Action, Comedy, Drama

2: Genre Filling (GF)

(b) Genre filling (GF)
Instruction: Below is a user’s interaction trail of movies he likes. Each movie is in double quotes “”, followed by its genre in single quotes ‘’. Your task is to fill in the genre placeholders represented by “?” with the following genres: [Action, Comedy, Drama]. The “_” represents placeholder tokens that you should not consider. The output should maintain the same format as the input.
Input: “Star Wars: Episode V - The Empire Strikes Back (1980)” ‘Action’, “Mission: Impossible (1996)” ‘Action’, ..., “Stargate (1994)” ‘Action’, “_” ‘?’, “_” ‘?’, ..., “_” ‘?’
Output: “Star Wars: Episode V - The Empire Strikes Back (1980)” ‘Action’, “Mission: Impossible (1996)” ‘Action’, ..., “Stargate (1994)” ‘Action’, “_” ‘Action’, “_” ‘Comedy’, ..., “_” ‘Comedy’

3: Item Predicting (IP)

(c) Item predicting (IP)
Instruction: Below is a user’s interaction trail of movies he likes. Each movie is in double quotes “”, followed by its genre in single quotes ‘’. Your task is to fill in the movie placeholders represented by “?”. The “_” represents placeholder tokens that you should not consider. The output should maintain the same format as the input.
Input: “Star Wars: Episode V - The Empire Strikes Back (1980)” ‘Action’, “Mission: Impossible (1996)” ‘Action’, ..., “Stargate (1994)” ‘Action’, “?” ‘Action’, “?” ‘Comedy’, ..., “?” ‘Comedy’
Output: “Star Wars: Episode V - The Empire Strikes Back (1980)” ‘Action’, “Mission: Impossible (1996)” ‘Action’, ..., “Stargate (1994)” ‘Action’, “Saving Private Ryan (1998)” ‘Action’, “Pretty Woman (1990)” ‘Comedy’, ..., “Ghost (1990)” ‘Comedy’

1: Genre Predicting (GP)

Dado el historial del usuario y el número de control N_c , predice **qué géneros deberían aparecer** en la lista futura.

Con $N_c = 3$

	Entrenamiento	Control
Input	Historial + "predice los géneros más probables"	Historial + "predice exactamente 3 géneros"
Output	"Acción, Drama, Comedia"	"Acción, Drama, Comedia"

2: Genre Filling (GF)

Toma el conjunto de géneros predichos por GP y los distribuye en las **posiciones futuras** de la lista, usando un formato fijo de placeholders

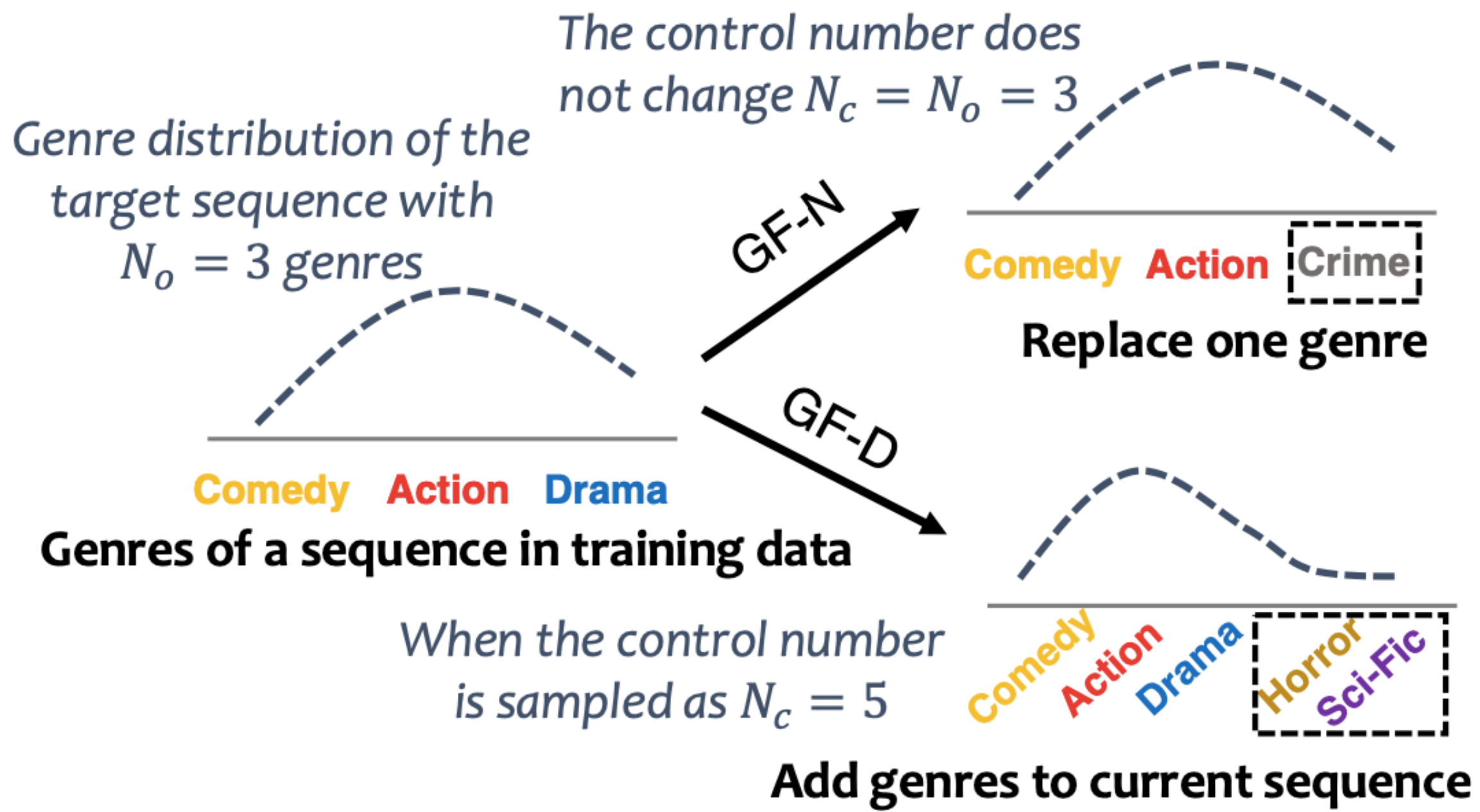
	Entrenamiento	Control
Input	Historial + placeholders + "usa géneros: [Acción, Drama, Comedia]"	Historial + placeholders + géneros de GP
Output	"Star Wars" 'Action', "Mission Impossible" 'Action', "Stargate" 'Action', "_" 'Action', "_" 'Drama', "_" 'Comedy'	Igual, con géneros propagados desde GP

3: Item Predicting (IP)

Predecir los **ítems concretos** que irán en cada posición de la lista final, guiado por los géneros asignados en GF

	Entrenamiento	Control
Input	Historial + "'?' 'Action', ' ?' 'Drama', '?' 'Comedy'"	Historial + lista de géneros de GF
Output	"Saving Private Ryan 'Action', Forrest Gump 'Drama', Pretty Woman 'Comedy'"	Lista final $\hat{L}$ con N_c géneros distintos

5. Detalle de la solución



Métodos

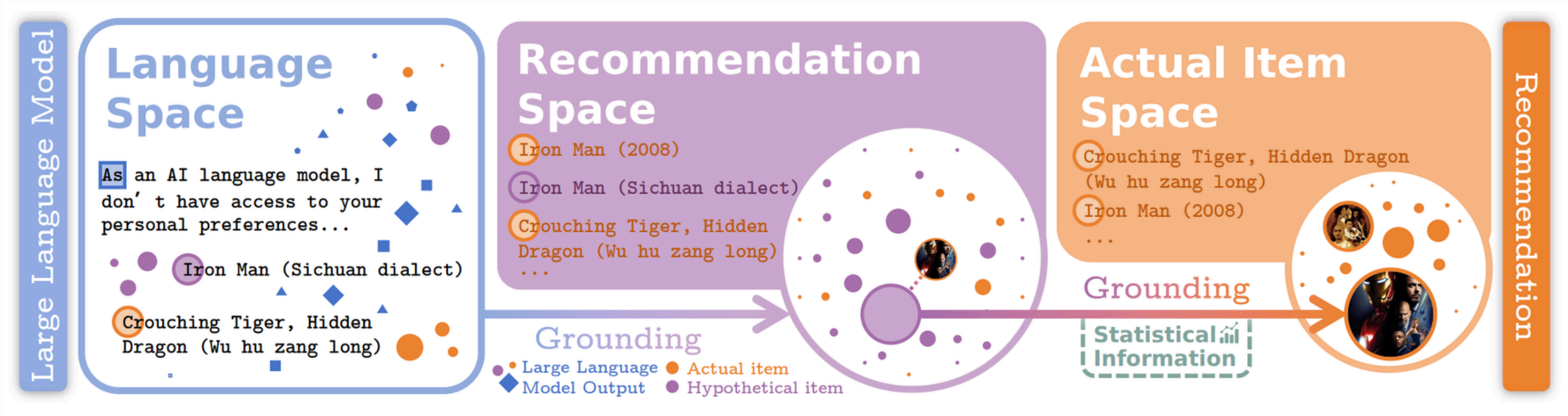
- BIGRec_div
- BIGRec_CoT
- DLCRecc

Métricas

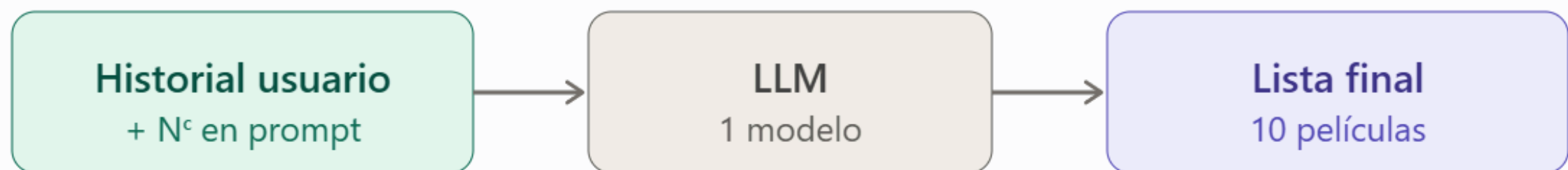
- NDCG@10
- Recall@10
- Cov@10
- MAE_Cov@10

¿Qué es BIGRec?

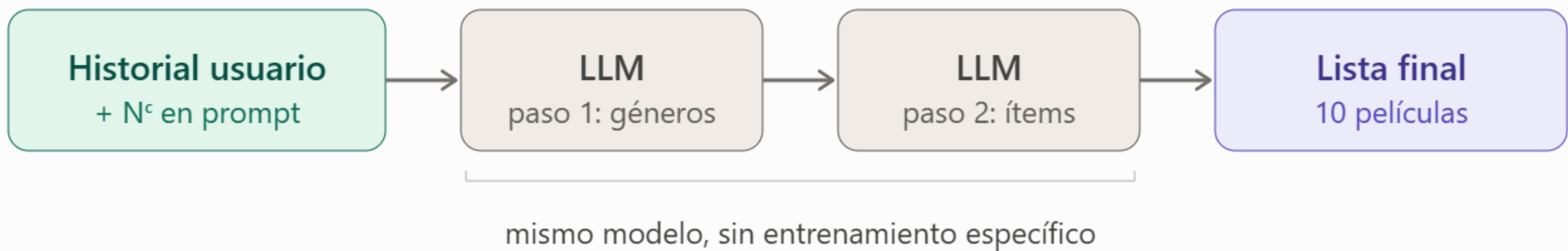
¿Qué es BIGRec?



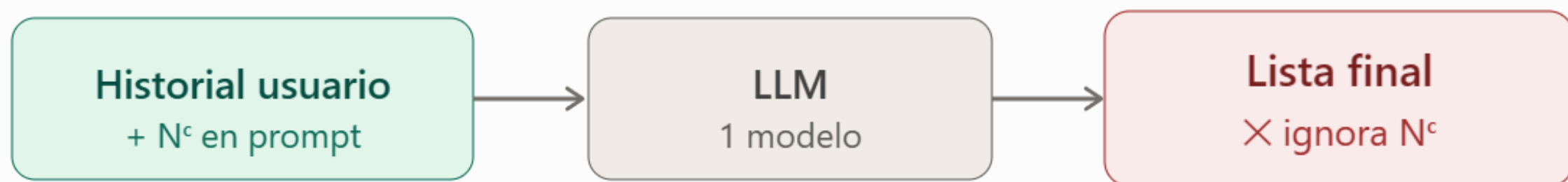
BIGRec_div



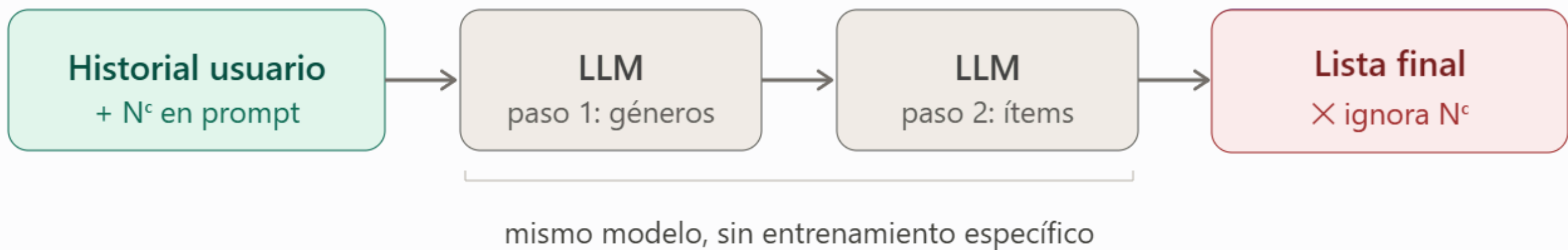
BIGRec_CoT (Chain-of-Thought)



BIGRec_div



BIGRec_CoT (Chain-of-Thought)



COV@10

Cobertura de géneros
en la lista
recomendada

MAE_Cov@10

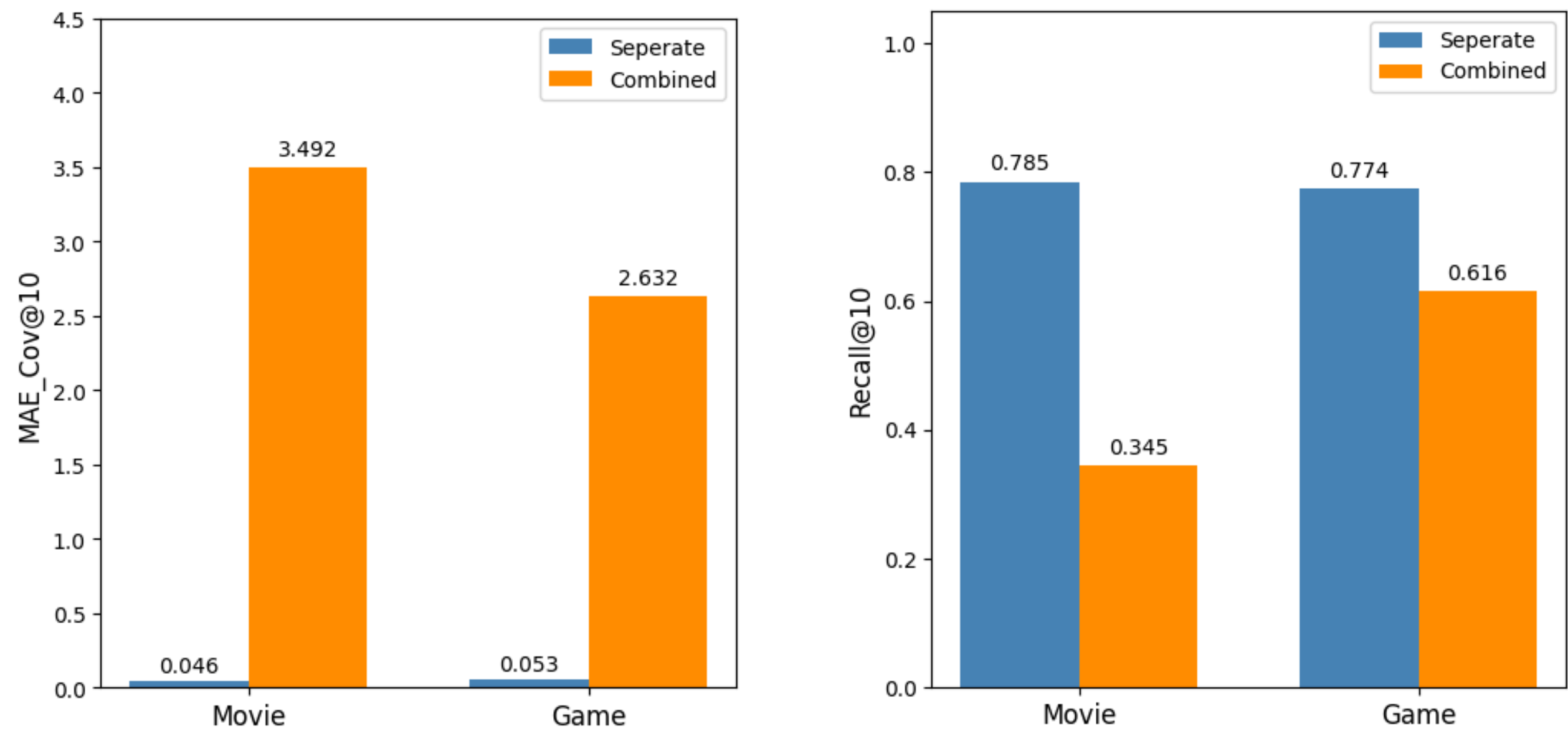
Error promedio
promedio entre la
cobertura de género
real y la deseada

6. Evaluación

Table 3: Performance comparison of DLCRec and other LLM-based baselines on Movie and Steam datasets. The best results in control metrics are highlighted in bold.

Method	Movie				Steam			
	NDCG@10	Recall@10	Cov@10	MAE_Cov@10	NDCG@10	Recall@10	Cov@10	MAE_Cov@10
control number=2								
BIGRec_div	0.0490	0.0428	2.526	1.112	0.0211	0.0173	2.689	0.827
BIGRec_CoT	0.0466	0.0396	2.517	0.987	0.0268	0.0218	2.242	0.486
DLCRec	0.0357	0.0306	2.478	0.798	0.0236	0.0201	1.655	0.445
control number=5								
BIGRec_div	0.0481	0.0426	2.515	2.529	0.0218	0.0182	2.713	2.291
BIGRec_CoT	0.0469	0.040	2.519	2.541	0.0277	0.0225	2.257	2.745
DLCRec	0.0450	0.0405	4.468	0.662	0.0289	0.0272	4.582	0.434
control number=8								
BIGRec_div	0.0480	0.0426	2.508	5.492	0.0216	0.0179	2.743	5.257
BIGRec_CoT	0.0470	0.0403	2.525	5.475	0.0274	0.0223	2.253	5.747
DLCRec	0.0436	0.0383	7.495	0.511	0.0273	0.0248	7.873	0.131

Figure 5: Performance comparison of separating and combining tasks GP and GF.

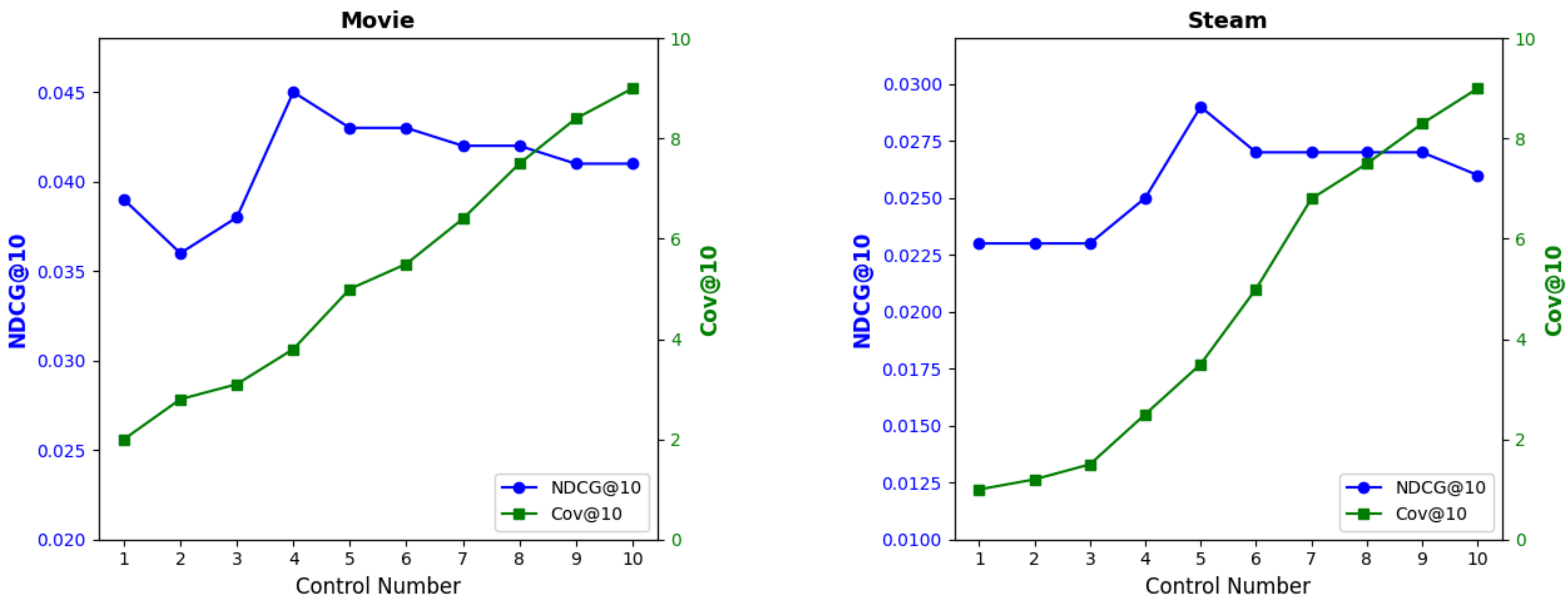


6. Evaluación

Table 4: Comparison of performances of DLCRec with different data augmentation strategies. "DA" denotes "Data Augmentation". "GF-ND" and "IP-ND" represent models using both data augmentation techniques, while "GF" and "IP" represent models without data augmentation. The best results in each group are highlighted in bold.

		Movie				Steam			
IP DA	GF DA	NDCG@10	Recall@10	COV@10	MAE_Cov@10	NDCG@10	Recall@10	COV@10	MAE_Cov@10
IP	GF	0.0390	0.0354	3.925	1.133	0.0274	0.0256	4.300	0.704
IP-ND	GF	0.0430	0.0396	4.426	0.710	0.0265	0.0249	4.529	0.485
IP-ND	GF-N	0.0437	0.0399	4.509	0.603	0.028	0.0262	4.524	0.494
IP-ND	GF-D	0.0425	0.0381	4.487	0.611	0.0260	0.0242	4.822	0.192
IP-ND	GF-ND	0.0450	0.0405	4.468	0.662	0.0289	0.0272	4.582	0.434
IP	GF-ND	0.0449	0.0416	3.927	1.109	0.0281	0.026	4.338	0.666
IP-N	GF-ND	0.0431	0.0390	4.296	0.740	0.0240	0.0214	4.514	0.492
IP-D	GF-ND	0.0454	0.0395	3.845	1.251	0.0282	0.0254	4.799	0.219
IP-ND	GF-ND	0.0450	0.0405	4.468	0.662	0.0289	0.0272	4.582	0.434

Figure 6: Sensitivity of DLCRec to varying levels of diversity



7. Conclusiones

- 1.- El **control de diversidad** funciona y no sacrifica precisión significativa.
- 2.- El **aumento de datos** fue lo que hizo posible que funcionara en los casos extremos.
- 3.- DLCRec no es un sistema completo, es una pieza que **resuelve algo que nadie había resuelto bien antes.**

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